

SELF-PLAY REINFORCEMENT LEARNING • IMPERFECT-INFORMATION GAMES

Knowing whether it was good was the hard half

*An AlphaHoldem-inspired self-play RL agent for Heads-Up No-Limit Hold'em —
and the evaluation discipline that kept catching my “better” models being worse.*

Sanil Desai • PyTorch • OpenSpiel • trained on a university GPU cluster

Building the agent was the easy half. Knowing whether it was good was the hard half.

● WHAT I BUILT

A complete, from-scratch deep-RL system for HUNL — PPO + GAE, a K-best self-play league, an FCPA→FCHPA curriculum, an automated eval harness, and a deployed play-against-it demo. All on a single GPU.

● THE HONEST RESULT

A balanced, gate-passing champion (entropy 1.44 bits). And an evaluation harness that repeatedly caught my “improved” models quietly regressing — when the training curves said they were winning.

● THE HONEST GAP

Not a Slumbot-beater, not near-Nash. Strong-but-exploitable, measured carefully. The deliverable is the measurement discipline — the value is the method, not a magnitude.

● THE SYSTEM

A complete, end-to-end self-play RL system — built from scratch in PyTorch.

● PSEUDO-SIAMESE NET

Two-tower actor-critic, ~1.5M params, search-free single forward pass.

● PPO + GAE

Clipped policy loss, Huber value, entropy regularization, KL-adaptive LR.

● K-BEST SELF-PLAY LEAGUE

Rolling snapshot pool, Elo-rated, prioritized opponent sampling.

● FCPA → FCHPA CURRICULUM

4- then 5-action betting abstractions over OpenSpiel universal_poker.

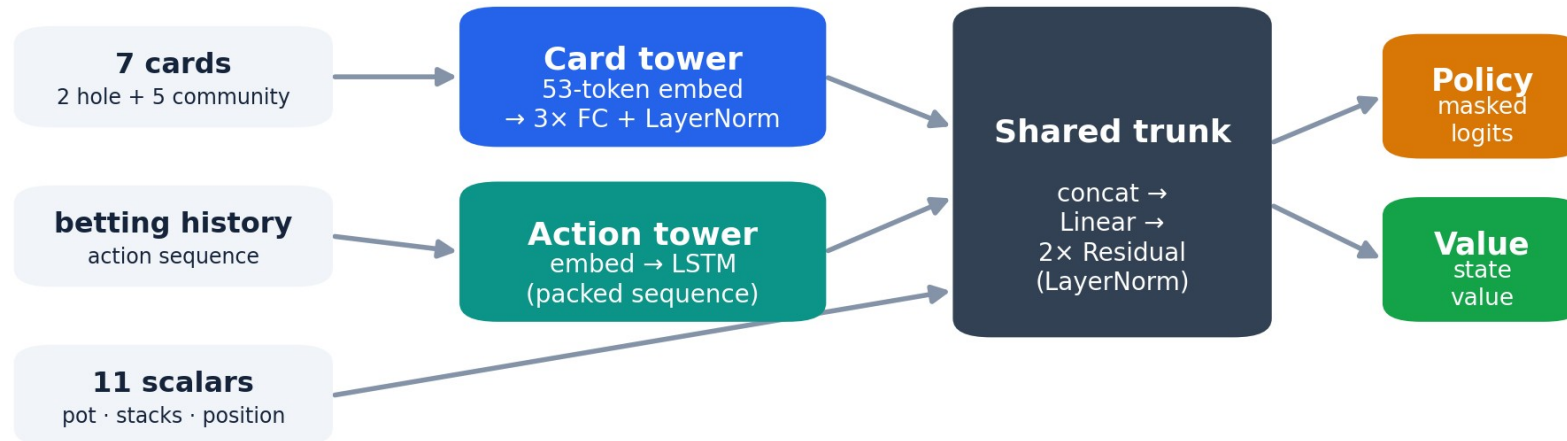
● EVALUATION HARNESS

Multi-seed Student-t CIs, behavior gates, duplicate mirrored-hand eval.

● SERVED & DEPLOYED

FastAPI inference backend + a browser demo you can play against.

A pseudo-siamese two-tower actor–critic — search-free, one masked forward pass per decision.



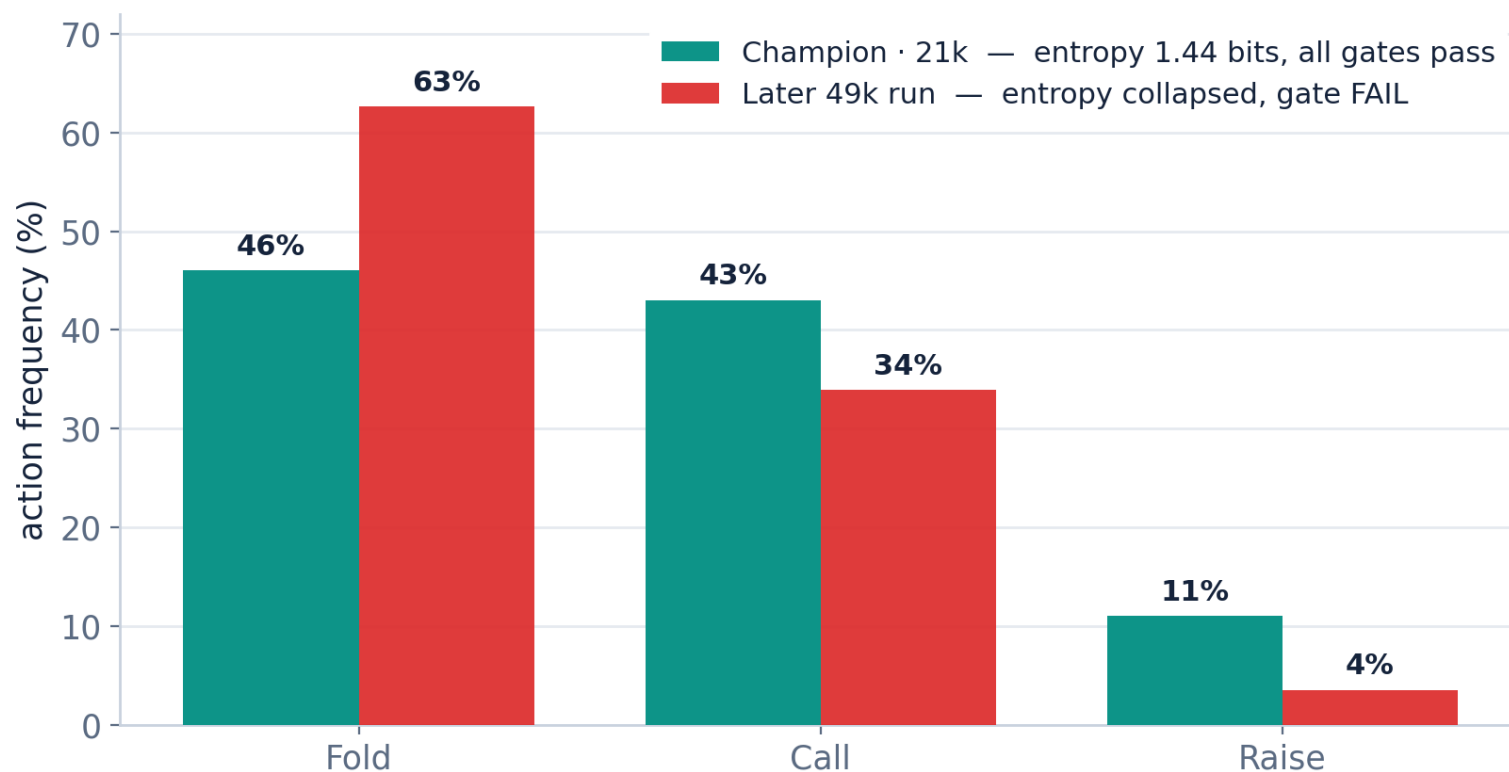
search-free: one masked forward pass per decision · trained with PPO + GAE against a K-best self-play league

This is AlphaHoldem-inspired — not a reproduction. I name every deviation.

Axis	AlphaHoldem (paper)	This repo
Card / action input	multi-channel tensors → ConvNets	53-token embedding → MLP / LSTM
Loss	Trinal-Clip PPO	vanilla clipped PPO + Huber value
Self-play	top-K by Elo	top-K rolling + rating-temperature sampling
Game scope	abstraction-free full game	F CPA (4) / F CHPA (5) betting abstraction
Inference	search-free, single forward pass	search-free, single forward pass ✓
Parameters	~8.6M	~1.5M

● DID IT ACTUALLY WORK?

The harness promotes on behavior gates, not win-rate — and it caught a policy collapse.



● A NUMBER I RETIRED

My flashiest number measured the opponent's weakness — not my agent's strength.

Champion win-rate vs the in-house “MCCFR-ES solver”, by solver strength:

+970

vs 0-iteration
uniform (random) bb/100

+898

vs 5k-iteration
“solver” bb/100

+934

vs 10k-iteration
“solver” bb/100

The 0-iteration (uniform-random) control scores the highest. Adding “solver” iterations does not make the opponent harder — the MCCFR-ES baseline is near-random on most states. So I retired “+897 bb/100 vs a solver,” and never place it next to AlphaHoldem’s +11 mbb/h vs Slumbot.

Win-rate is gameable; exploitability isn't — but on HUNL it's intractable to even measure.

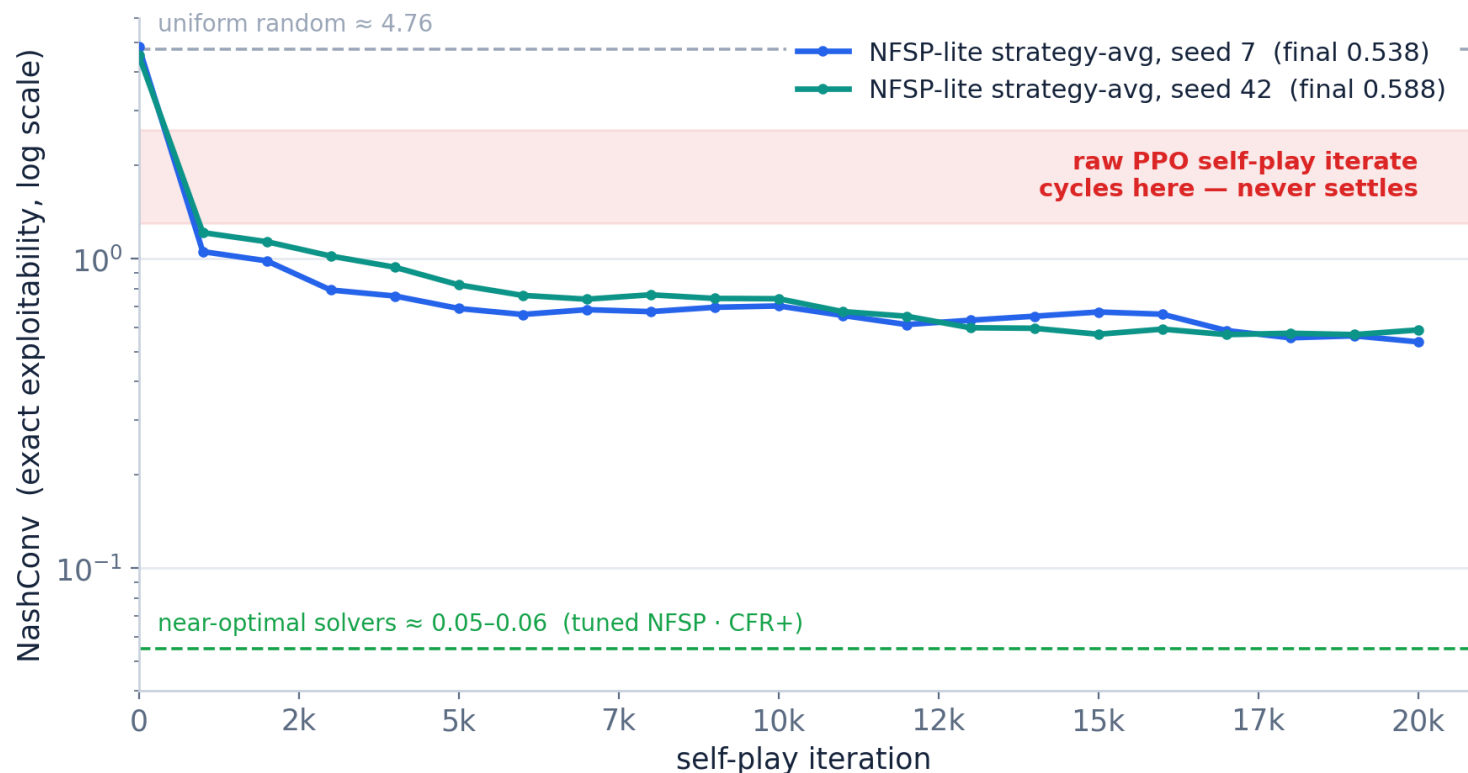
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information sets in full HUNL — exact best-response computation is prohibitive, so I make no exploitability claim about the HUNL agent.

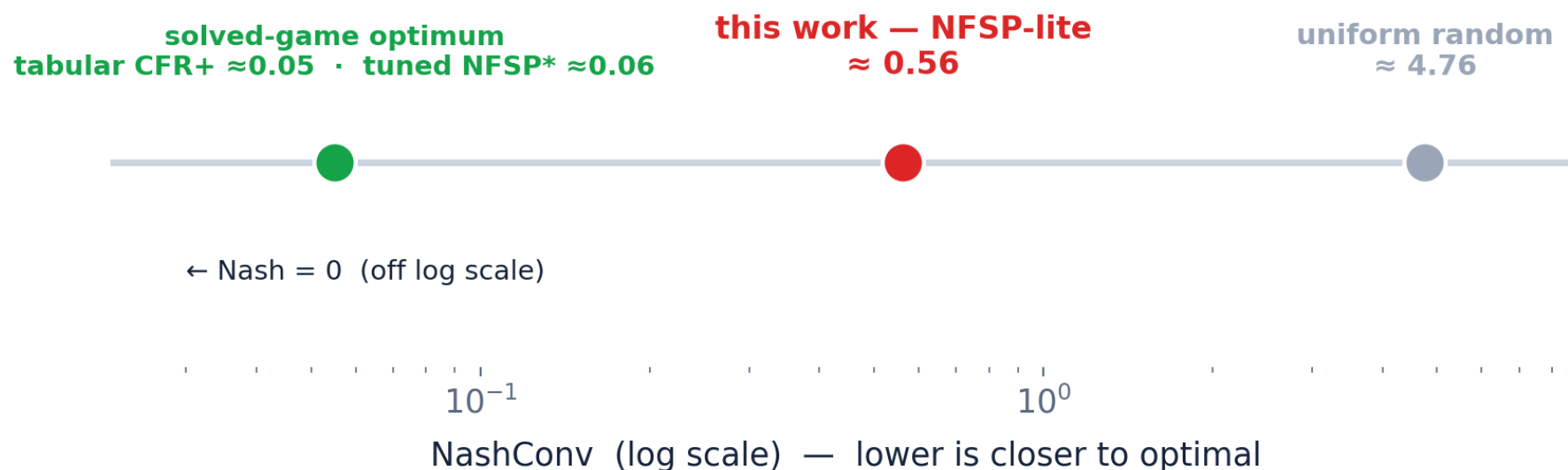
Lisý & Bowling (LBR): two bots tied head-to-head (within ~20 mbb/g) yet differed by ~1300 mbb/g in measured exploitability. A strong win-rate can hide a highly exploitable strategy.

So I moved the question to **Leduc poker** — 936 information states, where OpenSpiel computes exact NashConv — using the same PPO + self-play recipe as the HUNL configs.

On Leduc, where exploitability is exact: the raw PPO iterate cycles; only strategy-averaging is non-divergent.



The 0.56 plateau is methodology, not a headline — $\approx 8\times$ below random, but $\approx 9\times$ above tuned NFSP.



AlphaHoldem's “convergence” and my finding measure different quantities — there is no contradiction.

● WHAT ALPHAHOLDEM MEASURED

Training loss flattening · Elo plateauing in its self-play league · head-to-head win-rate (+111.6 mbb/h vs Slumbot). On full HUNL it explicitly concedes best-response is prohibitive and never computes a NashConv.

● WHAT EXPLOITABILITY MEASURES

The gain an optimal counter-strategy extracts — opponent-independent. As LBR shows, a strong head-to-head record is fully consistent with a strategy that remains highly exploitable.

Win-rate $\neq$ exploitability. My Leduc study is the test HUNL can't run — not a failed reproduction.

I made self-play collection 30–100× faster, then ran 90k GPU iterations from the champion.

30–100×

faster trajectory collection (batched inference vs one game at a time)

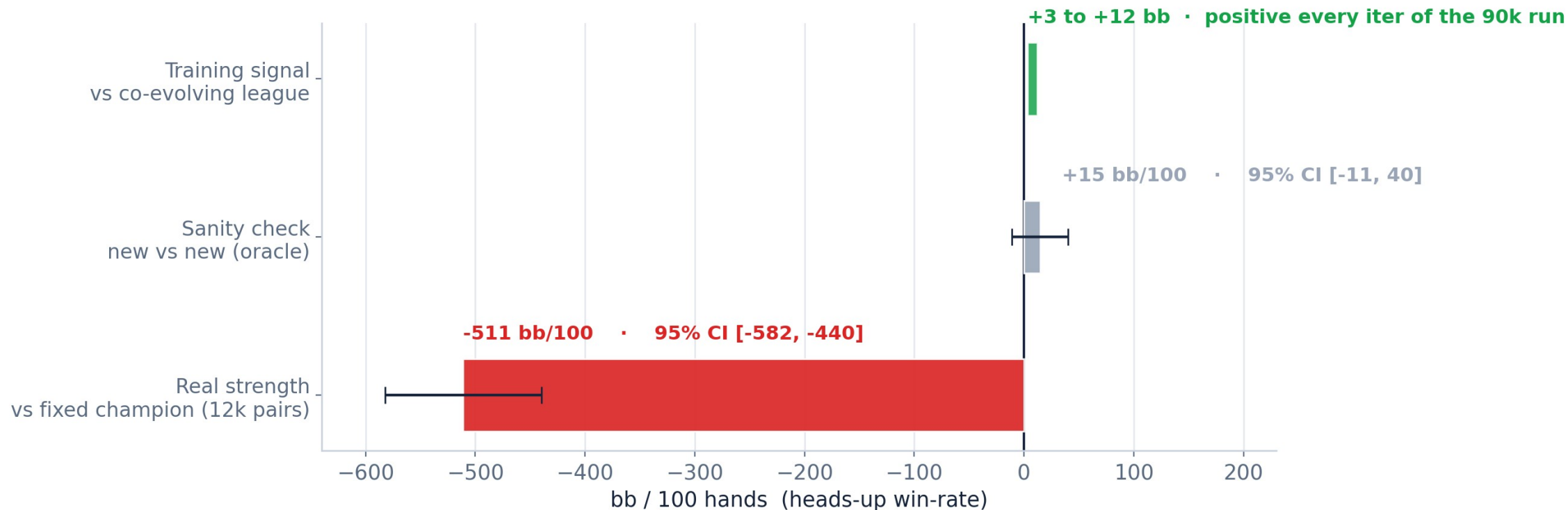
90k iters

~ 9×10^7 hands • ~38 h on one GPU

- **Rewrote the collector** to step B games in lockstep and batch all policy/value inference into a single forward pass — the GPU had been starved by Python-side game stepping.
- **Fixed an inverted PFSP bug** (opponent sampling had favored the strongest snapshots) and switched the loss to Trinal-Clip.
- **Warm-started from the Stage D 21k champion** and let it run — to see whether scale alone would push past a certified-good policy.

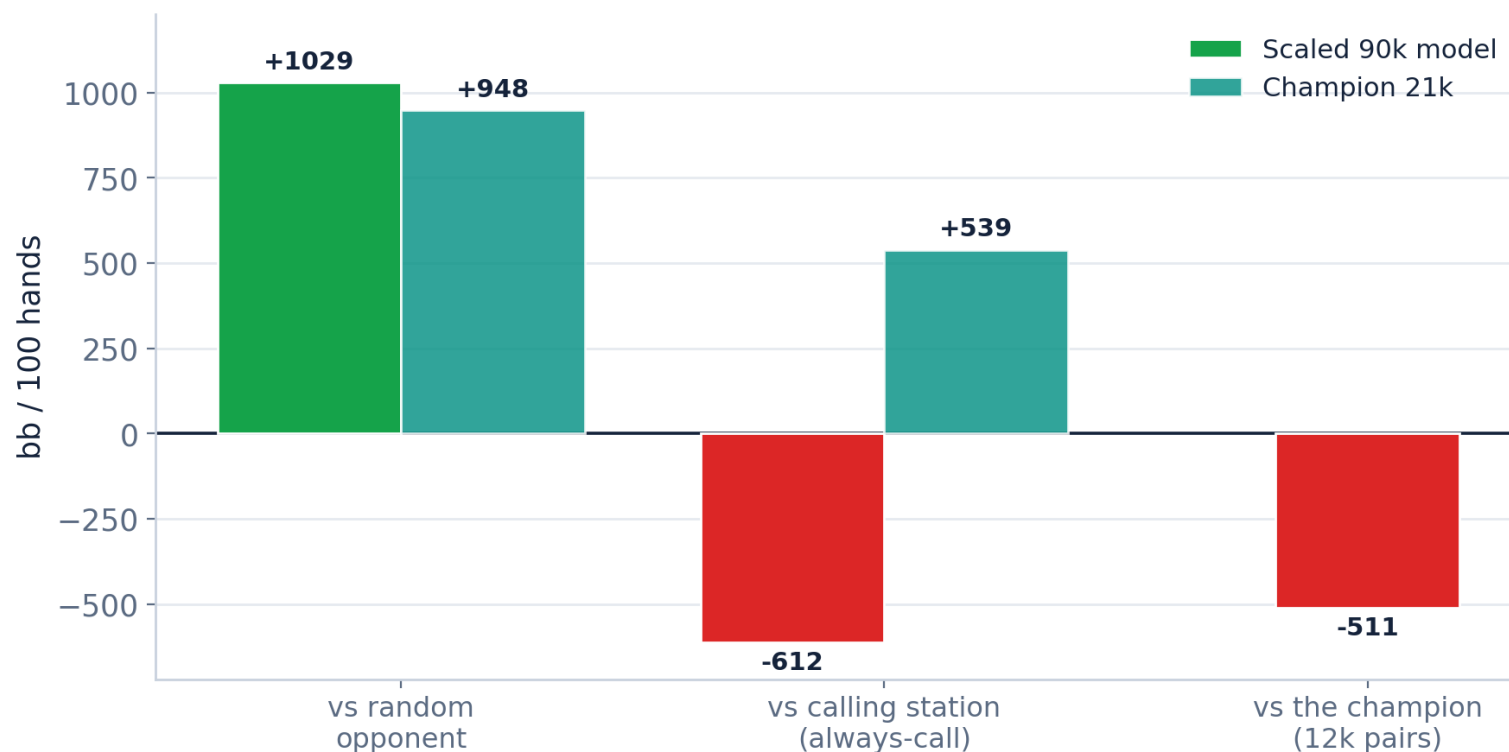
● THE REGRESSION THE EVALUATION CAUGHT

The training metric stayed positive the whole run; a fixed reference said it had regressed -511 bb/100.



● WHAT WENT WRONG

Over-aggression drift, not collapse: it crushes random even harder than the champion — but a calling station punishes it.



Choose ungameable metrics. Report honest negatives. Build the evaluation discipline to know when you're fooling yourself.

Ungameable metrics

Exact exploitability on Leduc, fixed-reference ladders, and an LBR lower bound — chosen because HUNL's metric is intractable.

Honest negatives

The iterate cycles, network-averaging fails, Trinal-Clip is inert at Leduc scale, the scaled run regressed -511 bb/100 — reported, not buried.

Evaluation discipline

The harness caught what the training curves structurally couldn't see — and refused to promote regressions automatically.

The honest ceiling: strong-but-exploitable, measured rigorously — not near-Nash, not a record win-rate. What transfers is first-principles systems building, measurement rigor, and the discipline to publish the negative result.